Candidate Seat no………….

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

**Sixth Semester of B. Tech (EC) Examination**

**April-May 2015**

**EC310 Optical Communication**

**Date: 09.05.2015, Saturday** **Time: 10:00 a.m. To 01.00 p.m.** **Maximum Marks: 70**

***Instructions:***

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

## SECTION – I

**Q - 1** **Answer the question below.** **[07]**

**a.** What is refractive index?

**b.** What makes fiber optic immune to EMI?

**c.** What is a skew ray?

**d.** Define quantum efficiency.

**e.** Which is an advantage of optical communication links over using transmission lines or waveguides?

a) Small size
b) Extremely wide bandwidth
c) Immune to EMI
d) All of above

**f.** The inner core of an optical fiber is _______ in composition.

a) Glass or Plastic
b) Copper
c) Bimetallic
d) Liquid

**g.** Dispersion is used to describe the

a) Splitting of light into its component colors.
b) Propagation of light in straight lines.
c) Bending of beam of light when its goes from one medium to another.
d) Bending of beam light when strikes a mirror.

**Q – 2.a** Consider a Multimode silica fiber that has a core refractive index $n_1$=1.48 and a cladding index $n_2$=1.46 calculate **[04]**

(a) Numeric Aperture NA (b) Acceptance angle

**Q – 2.b** **Answer any two questions.** **[10]**

**1)** Explain the key element of optical fiber communication system with diagram.

**2)** What is Numerical Aperture? Derive equation for the same.

**3)** What are the advantages of fiber optic?

**Q - 3** **Answer the following.**

**a.** Briefly discuss different type of losses in fiber optics. **[05]**

**b.**
1) Describe the construction of edge emitting LED with neat sketch. **[06]**
2) Explain lensing scheme to provide coupling between the source and diode **[03]**

**OR**

1) Explain what is meant by equilibrium NA. **[06]**
2) Explain three key transition processes involved in LASER diode. **[03]**

## SECTION – II

**Q - 4** **Answer the question below.** **[07]**

**a.** Define Response time of photo detector.

**b.** What is switching?

**c.** Which of the following is used as an optical receiver in fiber optics communication?

a) APD.
b) Tunnel Diode.
c) Laser Diode.
d) LED.

**d.** Define impact ionization.

**e.** SONET stands for ________________.

**f.** Define splitting ratio in case of optical coupler.

**g.** What is link power budget?

**Q – 5.a** An InGaAs *pin* photodiode has the following parameters at a wavelength of 1300 nm: $I_D$ = 4 nA, $\eta$ = 0.90, $R_L$ = 1000Ω, and the surface leakage current is negligible. the incident optical power is 300 nW(-35 dbm).Find the various noise term of the receiver for he bandwidth of 20 MHz .( $q = 1.6 * 10^{-19}$ c , $h = 6.625*10^{-34}$ Js , $c = 3 * 10^{8}$ m/s ) **[05]**

**OR**

**Q – 5.a** In One optical communication link have the photo diode which required receiver input signal of -42dbm and optical source that can transmit -13dbm average optical power level in to the fiber fly head with 50μm core diameter. Assume that 1db loss occur when the fiber fly head is connected to the cable and another 1db connector loss. Link also includes the 6db system margin. Find the possible transmission distant with $\alpha f$ = 3.5db/km **[05]**

**Q – 5.b** Explain quantum efficiency and responsivity of photodiode. **[04]**

**Q – 5.c** Explain principle of operation, the structure and the characteristics of the pin diode. **[05]**

**OR**

**Q – 5.c** Write a short note on photo detector noise. **[05]**

**Q – 6.a** What is Switching? Give the Difference Between Packet switching and Burst Switching. **[04]**

**Q – 6.b** Draw and Explain SONET frame structure. **[07]**

**Q – 6.c** What are the key system features of WDM? **[03]**

**OR**

**Q – 6.a** Explain 2 x 2 fiber coupler. What are the different losses associated with it? **[06]**

**Q – 6.b** Explain digital signal transmission in optical communication with figure. **[04]**

**Q – 6.c** Explain subcarrier multiplexing. **[04]**

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